

Harsh Pandey

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PROFESSIONAL SUMMARY

Final-year B.Tech (Information Technology) student who likes building systems end to end — training a YOLOv8 model, writing the REST API that serves its output, and getting the whole stack running in Docker. Comfortable across the stack: Java/Spring Boot and Node.js/Express on the backend, React on the frontend, FastAPI for AI-facing services, and SQL/NoSQL databases with CI/CD in between. Picks up new ecosystems fast — taught myself Swift/UIKit well enough to ship a small iOS app — and would rather trace a bug to its root cause than work around it. Looking for a full-stack, backend, or AI/ML engineering role.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, Kotlin, Swift

Backend: Node.js, Express.js, Spring Boot, Spring Security, FastAPI, REST APIs, JWT, MVC, Authentication, Authorization, RBAC

Frontend & Mobile: React.js, React Native, HTML5, CSS3, Android (Kotlin), Swift/UIKit

AI / ML: Machine Learning, Deep Learning, Computer Vision, Object Detection, YOLOv8, OpenCV, MediaPipe, LLM API Integration

Databases: MySQL, PostgreSQL, MongoDB, Firebase, Database Design, CRUD

DevOps & Tools: Docker, Docker Compose, Jenkins, CI/CD, Git/GitHub, Postman, Selenium

Blockchain: Solidity, Web3.py, Ethereum, Smart Contracts

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, System Design, Operating Systems, Computer Networks

EXPERIENCE

Web Development Intern — CodSoft (Virtual, Project-Based Internship) June 2025 – July 2025

- Completed full-stack web development projects as part of a virtual, project-based internship, connecting React frontends to Node.js/Express backends through REST APIs.
- Designed backend endpoints and data models for core application features, refining them through iterative testing.
- Debugged and validated APIs with Postman, managed version control with Git/GitHub, and deployed completed projects to Netlify.

PROJECTS

Intelligent Mob Surveillance System

Tech Used: YOLOv8 · OpenCV · FastAPI · MongoDB · Docker · Jenkins · Solidity · Web3.py

An AI-powered CCTV system that detects mob/crowd activity in real time and keeps the resulting video evidence verifiably intact.

- Built a real-time crowd/mob detection pipeline around YOLOv8 and OpenCV, exposed through a FastAPI backend with MongoDB for storing detections and metadata.
- Encrypted recorded evidence with AES-256 and hashed it with SHA-256 for data integrity, then anchored those hashes to an Ethereum blockchain via Solidity smart contracts and Web3.py for verifiable evidence integrity.
- Containerized the system as Docker microservices with a Jenkins CI/CD pipeline for automated builds and deployment.
- Deployed the stack to a cloud VM using Docker Compose, with HTTPS handled by Caddy.

SkillMatch

Tech Used: *Node.js · Express.js · MySQL · EJS*

A full-stack learning-recommendation application that suggests what a learner should study next based on their existing skills and target role.

- Structured the backend as an MVC application with session-based authentication (bcrypt) and role-based access control separating learner and admin capabilities.
- Built a recommendation engine that scores candidate skills by category match, target-role relevance, and difficulty to suggest a progressive learning path.
- Exposed the skill catalogue through a REST API with full CRUD operations and MySQL-backed data modeling, plus rate limiting and a health-check endpoint for operational reliability.

QuantumMind AI

Tech Used: *Spring Boot · FastAPI · React/Vite · PostgreSQL · Docker*

An AI-assisted application that answers questions grounded in stored data using retrieval-augmented generation, built as a set of coordinated backend services.

- Architected the app as four containerized services — a Spring Boot backend, a FastAPI AI service, a React/Vite frontend, and PostgreSQL — coordinated with Docker Compose.
- Implemented retrieval-augmented generation with FAISS vector similarity search over embedded content so the AI service grounds its answers in stored data rather than the model alone.
- Built the React/Vite frontend to consume streamed AI responses over Server-Sent Events (SSE) for incremental output.
- Designed the PostgreSQL schema and the REST integration between the Spring Boot and FastAPI services.

BrainMatch

Tech Used: *Swift · UIKit*

A native iOS memory-matching game — flip cards, find the pairs, beat your best time.

- Built the game logic and UI natively in Swift and UIKit, including card-flip animations and an in-app rules screen.
- Persisted game state with UserDefaults so a player can close the app mid-game and resume later.

ACHIEVEMENTS

- Winner, Best Idea Award — IdeaSpark 2K24, MIT-ADT University
- Participant — Smart India Hackathon (SIH)
- AMCAT Certified — quantitative, logical, and technical aptitude

CERTIFICATIONS

- Backend Development & API Creation — Coursera
- Node.js & MongoDB: Developing Backend — Coursera
- MongoDB Java Developer Path — MongoDB University
- Software Engineering & Agile Software Development — Infosys Springboard
- Cloud Technologies — Infosys Springboard
- Selenium Automation & Testing Frameworks — Coursera
- Database Management System — Infosys Springboard
- Networking Basics — Cisco / NPTEL

EDUCATION

B.Tech, Information Technology (Software & Mobile Application Development) — MIT-ADT University, Pune | 2023–2027 | CGPA: 8.38

HSC (Science) — Maharashtra State Board | 2022 | 63%

SSC — Maharashtra State Board | 2020 | 86.60%